

## CS-GY 9223 I: Lecture 11

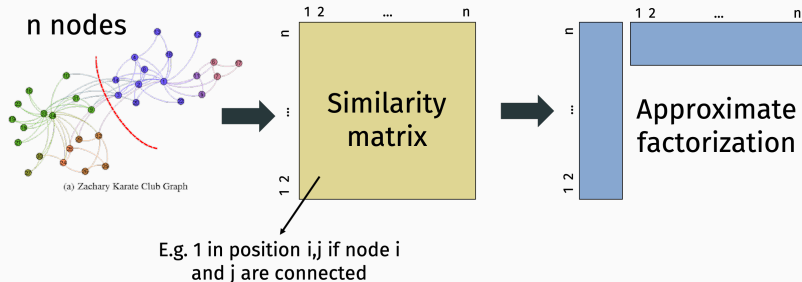
# Spectral graph theory + randomized numerical linear algebra.

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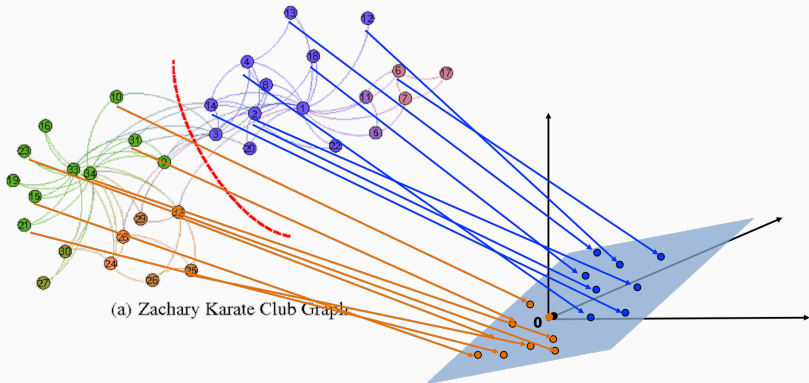
NYU Tandon School of Engineering, Prof. Christopher Musco

# ENCODING GRAPH SIMILARITY

Often data is represented as a graph and similarities can be obtained from that graph:



# ENCODING GRAPH SIMILARITY

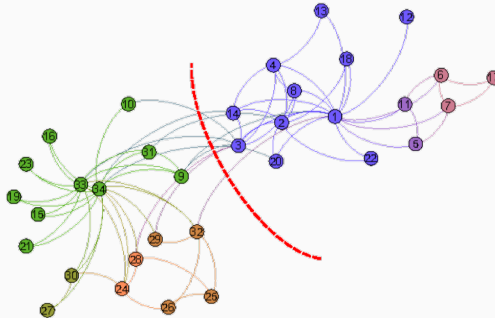


Spectral graph theory lets us formalize this heuristic idea.

# CUT MINIMIZATION

**Goal:** Partition nodes along a cut that:

- Has few crossing edges:  $|\{(u, v) \in E : u \in S, v \in T\}|$  is small.
- Separates large partitions:  $|S|, |T|$  are not too small.

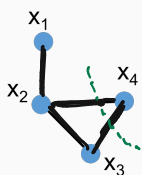


(a) Zachary Karate Club Graph

# THE LAPLACIAN VIEW

For a graph with adjacency matrix  $A$  and degree matrix  $D$ ,  
 $L = D - A$  is the **graph Laplacian**.

$$\|Bc\|_2^2 = \underbrace{c^T L c}_{= c^T B^T B c}$$



$$\begin{matrix} & \mathbf{D} & & \mathbf{A} & \\ \rightarrow & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} & - & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} & = & \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix} & = \mathbf{L} \end{matrix}$$

(cut indicator vector  $[1 \ 1 \ -1 \ -1]$ )

$L = B^T B$  where  $B$  is the "edge-vertex incidence" matrix.

|       |        |        |    |     |
|-------|--------|--------|----|-----|
|       | node 1 | node 2 | 3  | 4.. |
| edges | 1      | -1     | 0  | 0   |
|       | 0      | 1      | -1 | 0   |
|       | 0      | 1      | 0  | -1  |
|       | 0      | 0      | 1  | -1  |

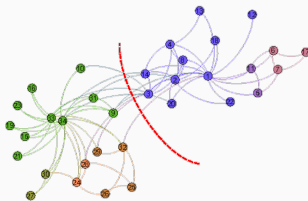
$$B = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$Bc = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 2 \end{bmatrix}$$

$Bc = \begin{cases} 0 & \text{for edges not across the cut} \\ 2 & \text{for edges across the cut} \end{cases}$

$$\|Bc\|_2^2 = 4 \cdot (\# \text{ edges across cut})^2 = 4 \cdot 5^2 = 100$$

# THE LAPLACIAN VIEW



(a) Zachary Karate Club Graph

For a cut indicator vector  $\mathbf{c} \in \{-1, 1\}^n$  with  $\mathbf{c}(i) = -1$  for  $i \in S$  and  $\mathbf{c}(i) = 1$  for  $i \in T$ :

- $\mathbf{c}^T \mathbf{L} \mathbf{c} = 4 \cdot \text{cut}(S, T)$ .
- $\mathbf{c}^T \mathbf{1} = |T| - |S|$ .  $\approx$  "imbalance in cut"

Want to minimize both  $\mathbf{c}^T \mathbf{L} \mathbf{c}$  (cut size) and  $\mathbf{c}^T \mathbf{1}$  (imbalance).

## Courant–Fischer min-max principle

Let  $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_n]$  be the eigenvectors of  $\mathbf{L}$ .

↓  
for symmetric  
matrix

$$\mathbf{v}_1 = \arg \max_{\|\mathbf{v}\|=1} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

$$\mathbf{v}_2 = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

$$\mathbf{v}_3 = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1, \mathbf{v}_2} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

⋮

$$\mathbf{v}_n = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1, \dots, \mathbf{v}_{n-1}} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

## Courant–Fischer min-max principle

Let  $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_n]$  be the eigenvectors of  $\mathbf{L}$ .

$$\mathbf{v}_n = \arg \min_{\|\mathbf{v}\|=1} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

$$\mathbf{v}_{n-1} = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

$$\mathbf{v}_{n-2} = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n, \mathbf{v}_{n-1}} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

$$\vdots$$

$$\mathbf{v}_1 = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n, \dots, \mathbf{v}_2} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

## SMALLEST LAPLACIAN EIGENVECTOR

The smallest eigenvector/singular vector  $\mathbf{v}_n$  satisfies:

$$\mathbf{v}_n = \frac{1}{\sqrt{n}} \cdot \mathbf{1} = \arg \min_{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

$\xrightarrow{\text{for } \mathbf{L}}$   $\frac{1}{\sqrt{n}} \cdot \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$

with  $\mathbf{v}_n^T \mathbf{L} \mathbf{v}_n = 0$ .

1) All of  $\mathbf{L}$ 's eigenvalues are  $\geq 0$ .

$\mathbf{L}$  is positive semidefinite:  $\mathbf{B}^T \mathbf{B} = \mathbf{L}$ .

$\mathbf{B}^T \mathbf{B}$

$$2) \mathbf{L} \cdot \mathbf{1} = \mathbf{B}^T \mathbf{B} \cdot \mathbf{1} = \mathbf{B}^T \mathbf{0} = \mathbf{0}$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \mathbf{1} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\mathbf{L} = \left( \frac{1}{\sqrt{n}} \mathbf{1} \right) \left( \frac{1}{\sqrt{n}} \mathbf{1} \right)^T = \mathbf{0} \left( \frac{1}{\sqrt{n}} \mathbf{1} \right)^T$$

## SECOND SMALLEST LAPLACIAN EIGENVECTOR

By Courant-Fischer,  $\mathbf{v}_{n-1}$  is given by:

$$\mathbf{v}_{n-1} = \underset{\|\mathbf{v}\|=1, \mathbf{v}_n^T \mathbf{v}=0}{\operatorname{argmin}} \mathbf{v}^T \mathbf{L} \mathbf{v}$$

$$\mathbf{v}_n = \frac{1}{\sqrt{n}} \cdot \mathbf{1}$$

cut value for  
cut indicators  
add constraint  
 $\mathbf{v}_{n-1} \in \{ \frac{1}{\sqrt{n}}, -\frac{1}{\sqrt{n}} \}$

If  $\mathbf{v}_{n-1}$  were binary, i.e.  $\in \{-1, 1\}^n$ , scaled by  $\frac{1}{\sqrt{n}}$ , it would have:

- $\mathbf{v}_{n-1}^T \mathbf{L} \mathbf{v}_{n-1} = \text{cut}(S, T)$  as small as possible **given that**  
 **$\mathbf{v}_{n-1}^T \mathbf{1} = |T| - |S| = 0$**
- $\mathbf{v}_{n-1}$  would indicate the smallest perfectly balanced cut.

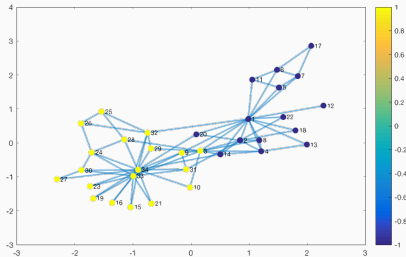
$\mathbf{v}_{n-1} \in \mathbb{R}^n$  is not generally binary, but still satisfies a 'relaxed' version of this property.

## CUTTING WITH THE SECOND LAPLACIAN EIGENVECTOR

Find a good partition of the graph by using an eigendecomposition to compute

$$\mathbf{v}_{n-1} = \arg \min_{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1, \mathbf{v}^T \mathbf{1}=0} \mathbf{v}^T L \mathbf{v}$$

Set  $S$  to be all nodes with  $\mathbf{v}_{n-1}(i) < 0$ , and  $T$  to be all with  $\mathbf{v}_{n-1}(i) \geq 0$ .



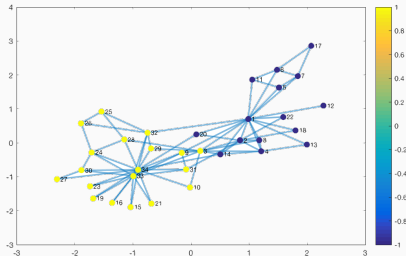
$\mathbf{v}_{n-1}$  :

## CUTTING WITH THE SECOND LAPLACIAN EIGENVECTOR

Find a good partition of the graph by using an eigendecomposition to compute

$$\mathbf{v}_{n-1} = \underset{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1, \mathbf{v}^T \mathbf{1} = 0}{\operatorname{arg\,min}} \quad \mathbf{v}^T L \mathbf{v}$$

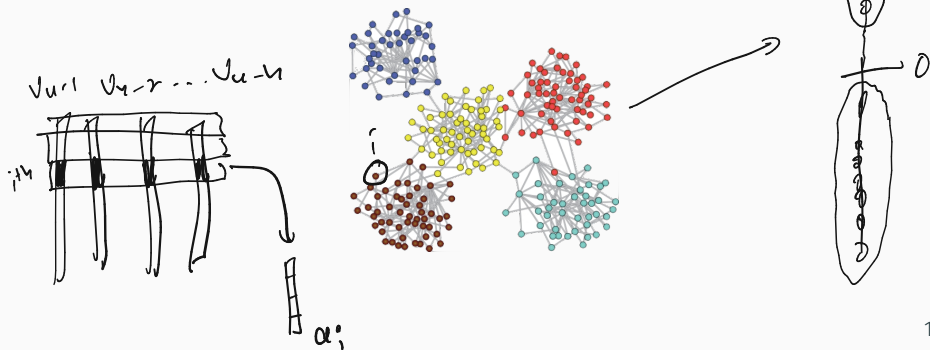
Set  $S$  to be all nodes with  $\mathbf{v}_{n-1}(i) < 0$ , and  $T$  to be all with  $\mathbf{v}_{n-1}(i) \geq 0$ .



## SPECTRAL PARTITIONING IN PRACTICE

The Shi-Malik normalized cuts algorithm is one of the most commonly used variants of this approach, using the normalized Laplacian  $\bar{\mathbf{L}} = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$ .

**Important consideration:** What to do when we want to split the graph into more than two parts?



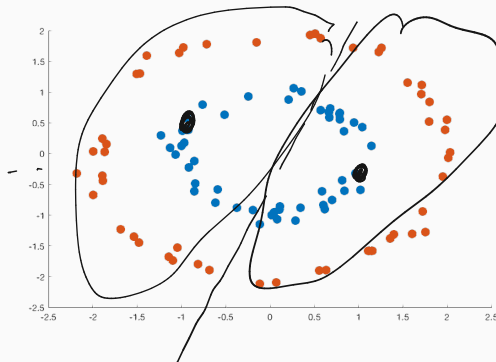
### Spectral Clustering:

- Compute smallest  $k$  eigenvectors  $\mathbf{v}_{n-1}, \dots, \mathbf{v}_{n-k}$  of  $\mathbf{L}$ .
- Represent each node by its corresponding row in  $\mathbf{V} \in \mathbb{R}^{n \times k}$  whose rows are  $\mathbf{v}_{n-1}, \dots, \mathbf{v}_{n-k}$ .
- Cluster these rows using  $k$ -means clustering (or really any clustering method).

# LAPLACIAN EMBEDDING

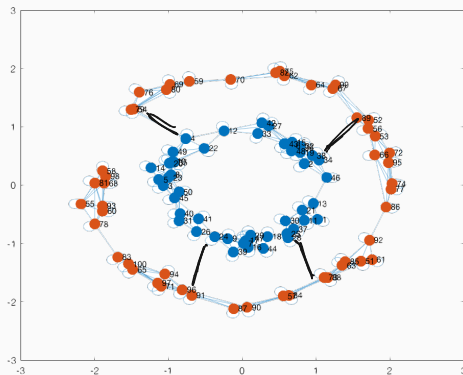
*2-means*

Original Data: (not linearly separable)

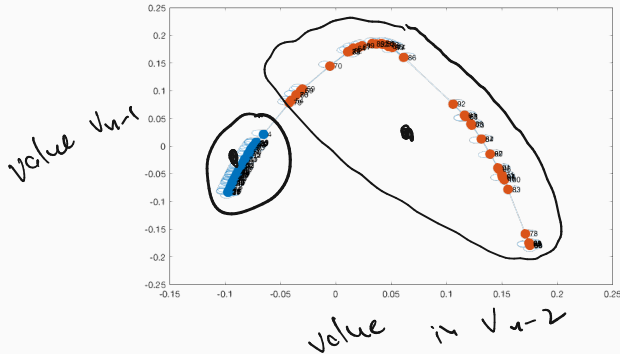


# LAPLACIAN EMBEDDING

↗ 5 6 10  
 $k$ -Nearest Neighbors Graph:



Embedding with eigenvectors  $\mathbf{v}_{n-1}, \mathbf{v}_{n-2}$ : (linearly separable)



## GENERATIVE MODELS

**So far:** Spectral clustering partitions a graph along a small cut between large pieces.

- No formal guarantee on the 'quality' of the partitioning.
- Would be difficult to analyze for general input graphs.

**Common approach:** Give a natural **generative model** for which produces random but realistic inputs and analyze how the algorithm performs on inputs drawn from this model.

- Very common in algorithm design for data analysis/machine learning (can be used to justify  $\ell_2$  linear regression,  $k$ -means clustering, PCA, etc.)

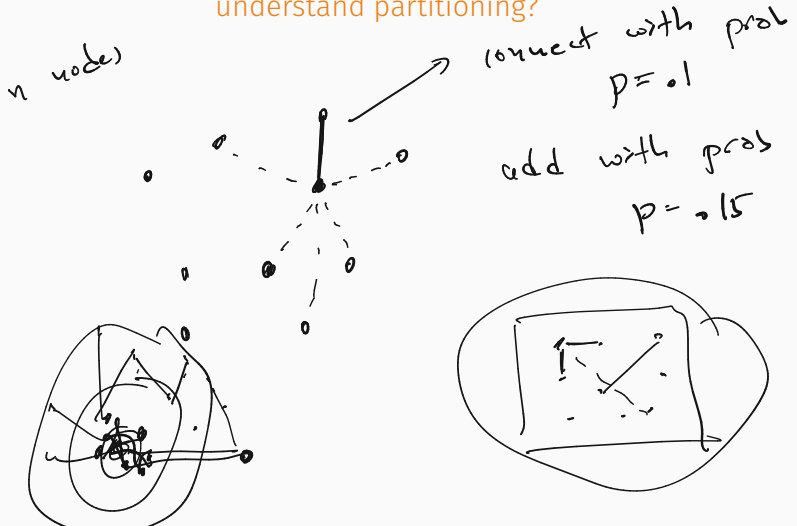
$$a_i^T x = y_i$$

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

$$x = \min \|Ax - b\|_{\ell_1, \ell_0}$$

# STOCHASTIC BLOCK MODEL

Ideas for a generative model for graphs that would allow us to understand partitioning?



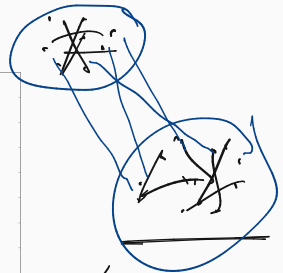
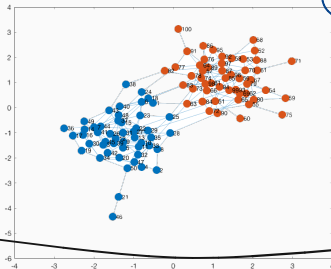
# STOCHASTIC BLOCK MODEL

## Stochastic Block Model (Planted Partition Model):

Let  $G_n(p, q)$  be a distribution over graphs on  $n$  nodes, split equally into two groups  $\underline{B}$  and  $\underline{C}$ , each with  $\underline{n/2}$  nodes.

- Any two nodes in the **same group** are connected with probability  $\underline{p}$  (including self-loops).  $p = 0.1$
- Any two nodes in **different groups** are connected with prob.  $q < p$ .

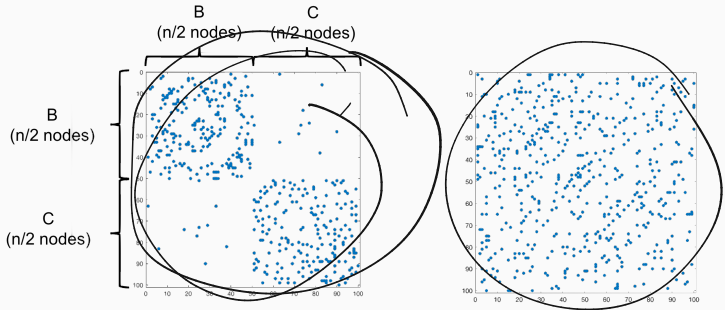
Erdos-Renyi  
 $G_{n/2}(p)$



# LINEAR ALGEBRAIC VIEW

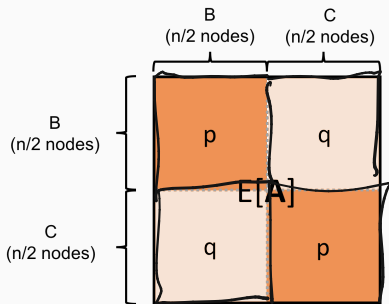
Let  $G$  be a stochastic block model graph drawn from  $G_n(p, q)$ .

- Let  $\mathbf{A} \in \mathbb{R}^{n \times n}$  be the adjacency matrix of  $G$ . What is  $\mathbb{E}[\mathbf{A}]$ ?



## EXPECTED ADJACENCY SPECTRUM

Letting  $G$  be a stochastic block model graph drawn from  $G_n(p, q)$  and  $\mathbf{A} \in \mathbb{R}^{n \times n}$  be its adjacency matrix.  $(\mathbb{E}[\mathbf{A}])_{i,j} = p$  for  $i, j$  in same group,  $(\mathbb{E}[\mathbf{A}])_{i,j} = q$  otherwise.



What are the eigenvectors and eigenvalues of  $\mathbb{E}[\mathbf{A}]$ ?

## EXPECTED ADJACENCY SPECTRUM

Letting  $G$  be a stochastic block model graph drawn from  $G_n(p, q)$  and  $A \in \mathbb{R}^{n \times n}$  be its adjacency matrix, what are the eigenvectors and eigenvalues of  $\mathbb{E}[A]$ ?

$$\mathbb{E}[A] = \begin{bmatrix} \text{---} & \text{---} \\ p & q \\ \text{---} & \text{---} \\ q & p \\ \text{---} & \text{---} \end{bmatrix}$$

$$\mathbb{E}[A] \cdot \mathbf{1} =$$

$$\begin{bmatrix} \frac{(p+q)n}{2} \\ \vdots \\ \frac{(p+q)n}{2} \\ \vdots \\ \frac{(p+q)n}{2} \end{bmatrix} = \frac{(p+q)n}{2} \mathbf{1}$$

$\underbrace{\hspace{1cm}}_1$

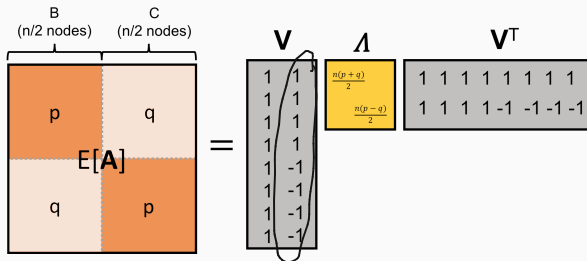
Eigenvector 1:  $\frac{1}{\sqrt{n}} \mathbf{1}$

Eigenvalue 1:  $\frac{(p+q)n}{2}$

Eigenvector 2:  $\frac{1}{\sqrt{n}} \begin{bmatrix} 1 \\ \vdots \\ 1 \\ \vdots \\ 1 \end{bmatrix} \rightarrow u$

$$\mathbb{E}[A] \cdot u = \begin{bmatrix} (p-q)n/2 \\ \vdots \\ (p-q)n/2 \\ \vdots \\ (q-p)n/2 \end{bmatrix} = \frac{(p-q)n}{2} u$$

## EXPECTED ADJACENCY SPECTRUM



- $\mathbf{v}_1 = \mathbf{v}_1$  with eigenvalue  $\lambda_1 = \frac{(p+q)n}{2}$ .
- $\mathbf{v}_2 = \underline{\chi_{B,C}}$  with eigenvalue  $\lambda_2 = \frac{(p-q)n}{2}$ .
- $\chi_{B,C}(i) = 1$  if  $i \in B$  and  $\chi_{B,C}(i) = -1$  for  $i \in C$ .

If we compute  $\mathbf{v}_2$  then we recover the communities  $B$  and  $C$ !

## EXPECTED LAPLACIAN SPECTRUM

Letting  $G$  be a stochastic block model graph drawn from  $G_n(p, q)$ ,  $\mathbf{A} \in \mathbb{R}^{n \times n}$  be its adjacency matrix and  $\mathbf{L}$  be its Laplacian, what are the eigenvectors and eigenvalues of  $\mathbb{E}[\mathbf{L}]$ ?

**Upshot:** The second small eigenvector of  $\mathbb{E}[\mathbf{L}]$  is  $\chi_{B,C}$  – the indicator vector for the cut between the communities.

- If the random graph  $G$  (equivalently  $\mathbf{A}$  and  $\mathbf{L}$ ) were exactly equal to its expectation, partitioning using this eigenvector would exactly recover communities  $B$  and  $C$ .

How do we show that a matrix (e.g.,  $\mathbf{A}$ ) is close to its expectation? **Matrix concentration inequalities.**

- Analogous to scalar concentration inequalities like Markov's, Chebyshev's, Bernstein's.

**Matrix Concentration Inequality:** If  $p \geq O\left(\frac{\log^4 n}{n}\right)$ , then with high probability

$$\|A - \mathbb{E}[A]\|_2 \leq O(\sqrt{pn}).$$

where  $\|\cdot\|_2$  is the matrix **spectral** norm (operator norm).

For  $X \in \mathbb{R}^{n \times d}$ ,  $\|X\|_2 = \max_{z \in \mathbb{R}^d: \|z\|_2=1} \|Xz\|_2$ .  $\max (z^T A z)$

**Exercise:** Show that  $\|X\|_2$  is equal to the largest singular value of  $X$ . For symmetric  $X$  (like  $A - \mathbb{E}[A]$ ) show that it is equal to the magnitude of the largest magnitude eigenvalue.

For the stochastic block model application, we want to show that the second eigenvectors of  $A$  and  $\mathbb{E}[A]$  are close. How does this relate to their difference in spectral norm?

# EIGENVECTOR PERTURBATION

1974

**Davis-Kahan Eigenvector Perturbation Theorem:** Suppose  $\mathbf{A}, \bar{\mathbf{A}} \in \mathbb{R}^{d \times d}$  are symmetric with  $\|\mathbf{A} - \bar{\mathbf{A}}\|_2 \leq \epsilon$  and eigenvectors  $\underline{v_1, v_2, \dots, v_d}$  and  $\underline{\bar{v}_1, \bar{v}_2, \dots, \bar{v}_d}$ . Letting  $\theta(v_i, \bar{v}_i)$  denote the angle between  $v_i$  and  $\bar{v}_i$ , for all  $i$ :

measure of closeness between  $v_i, \bar{v}_i$

$$\sin[\theta(v_i, \bar{v}_i)] \leq \frac{\epsilon}{\min_{j \neq i} |\lambda_i - \lambda_j|}$$

where  $\lambda_1, \dots, \lambda_d$  are the eigenvalues of  $\bar{\mathbf{A}}$ .

The error gets larger if there are eigenvalues with similar magnitudes.

# EIGENVECTOR PERTURBATION

$$\begin{array}{ccc} \mathbf{A} & \mathbf{\bar{A}} & \mathbf{A} - \mathbf{\bar{A}} \\ \begin{bmatrix} 1+\varepsilon & 0 \\ 0 & 1 \end{bmatrix} & - \begin{bmatrix} 1 & 0 \\ 0 & 1+\varepsilon \end{bmatrix} & = \begin{bmatrix} \varepsilon & 0 \\ 0 & \varepsilon \end{bmatrix} \\ \downarrow & & \\ \begin{array}{cc} v_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} & v_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \lambda_1 = 1+\varepsilon & \lambda_2 = 1 \end{array} & \begin{array}{cc} \begin{bmatrix} 0 \\ 1 \end{bmatrix} & \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \lambda_1 = 1+\varepsilon & \lambda_2 = 1 \end{array} & \begin{array}{c} \|A - \bar{A}\|_L = \varepsilon \end{array} \end{array}$$

# APPLICATION TO STOCHASTIC BLOCK MODEL

Claim 1 (Matrix Concentration): For  $p \geq O\left(\frac{\log^4 n}{n}\right)$ ,

$$\|A - \mathbb{E}[A]\|_2 \leq O(\sqrt{pn}).$$

Claim 2 (Davis-Kahan): For  $p \geq O\left(\frac{\log^4 n}{n}\right)$ ,

$$\sin \theta(v_2, \bar{v}_2) \leq \frac{O(\sqrt{pn})}{\min_{j \neq 2} |\lambda_2 - \lambda_j|} \leq \frac{O(\sqrt{pn})}{(p-q)n/2} = O\left(\frac{\sqrt{p}}{(p-q)\sqrt{n}}\right)$$

Recall:  $\mathbb{E}[A]$ , has eigenvalues  $\lambda_1 = \frac{(p+q)n}{2}$ ,  $\lambda_2 = \frac{(p-q)n}{2}$ ,  $\lambda_i = 0$  for  $i \geq 3$ .

$$\min_{j \neq i} |\lambda_i - \lambda_j| = \min\left(qn, \frac{(p-q)n}{2}\right).$$

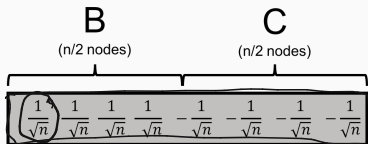
Assume  $\frac{(p-q)n}{2}$  will be the minimum of these two gaps.

$$\frac{(p+q)n}{2} \quad \frac{(p-q)n}{2} \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \frac{(p-q)n}{2}$$

## APPLICATION TO STOCHASTIC BLOCK MODEL

So Far:  $\sin \theta(v_2, \bar{v}_2) \leq O\left(\frac{\sqrt{p}}{(p-q)\sqrt{n}}\right)$ . What does this give us?

- Can show that this implies  $\|v_2 - \bar{v}_2\|_2^2 \leq O\left(\frac{p}{(p-q)^2 n}\right)$  (exercise).
- $\bar{v}_2$  is  $\frac{1}{\sqrt{n}} \chi_{B,C}$ : the community indicator vector.



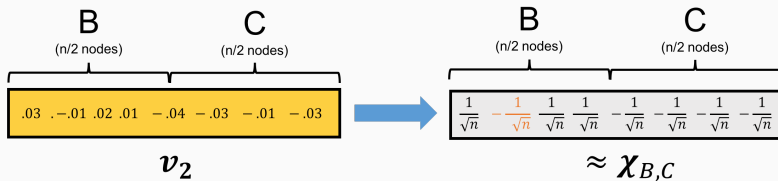
$v_2$

$\bar{v}_2 = [0.1, 0.2, -0.001, -0.1, -0.5]$

- Every  $i$  where  $v_2(i), \bar{v}_2(i)$  differ in sign contributes  $\geq \frac{1}{n}$  to  $\|v_2 - \bar{v}_2\|_2^2$ .
- So they differ in sign in at most  $O\left(\frac{p}{(p-q)^2}\right)$  positions.

## APPLICATION TO STOCHASTIC BLOCK MODEL

**Upshot:** If  $G$  is a stochastic block model graph with adjacency matrix  $A$ , if we compute its second large eigenvector  $v_2$  and assign nodes to communities according to the sign pattern of this vector, we will correctly assign all but  $O\left(\frac{p}{(p-q)^2}\right)$  nodes.



- Why does the error increase as  $q$  gets close to  $p$ ?
- Even when  $p - q = O(1/\sqrt{n})$ , assign all but an  $O(n)$  fraction of nodes correctly. E.g., assign 99% of nodes correctly.

$$p - q = O(1/\sqrt{n}) \Rightarrow \frac{p}{(p-q)^2} = O(n) \sim$$

# RANDOMIZED NUMERICAL LINEAR ALGEBRA

Forget about the previous problem, but still consider the matrix  $\mathbf{M} = \mathbb{E}[\mathbf{A}]$ .

- Dense  $n \times n$  matrix.
- Computing top eigenvectors takes  $\approx O(n^2/\sqrt{\epsilon})$  time.

If someone asked you to speed this up and return approximate top eigenvectors, what could you do?.

Hand-drawn diagram of a 2D array with 2 rows and 3 columns. The first row contains the values 1, 1, 1. The second row contains the values 1, 2, 1. An arrow points from the text  $[0,1]$  to the first element of the first row (the value 1). Below the array, the text  $= \{A\}$  is written.

$$\begin{array}{r} 10001 \\ 1000 \\ \hline 0001000 \end{array}$$

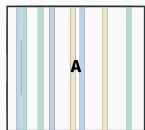
# RANDOMIZED NUMERICAL LINEAR ALGEBRA

**Main idea:** If you want to compute singular vectors or eigenvectors, multiply two matrices, solve a regression problem, etc.:

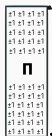
1. Compress your matrices using a randomized method.
2. Solve the problem on the smaller or sparser matrix.

•  $\tilde{A}$  called a “sketch” or “coreset” for  $A$ .

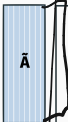
(column sampling)



compute  
approximate  
left singular  
vectors

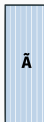
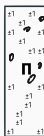


random  
matrix



sampling

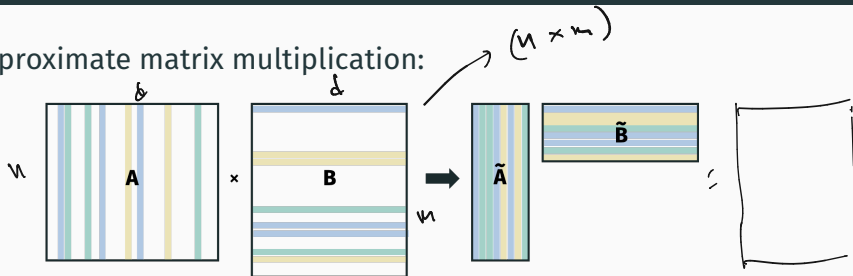
sparse  
matrix



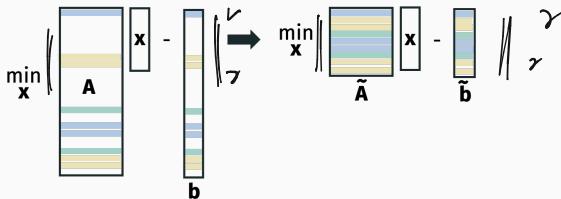
$$AA^T$$
$$\tilde{A}\tilde{A}^T$$
$$\tilde{A}A^T$$

# RANDOMIZED NUMERICAL LINEAR ALGEBRA

Approximate matrix multiplication:



Approximate regression:

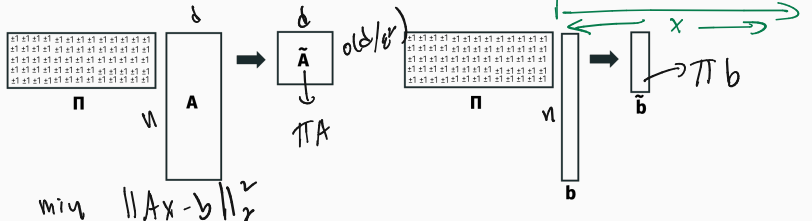


# COMPARISON

|           | Direct<br>Gaussian Elimination<br>QR Algorithms     | Iterative<br>Gradient Descent<br>+ Power Method<br>+ Krylov Methods                        | Randomized<br>sketch<br>↓<br>sampling,<br>random projection      |
|-----------|-----------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Method:   |                                                     |                                                                                            |                                                                  |
| Speed:    | $O(n^3)$ for<br>inverse, eigendecomp<br><b>Slow</b> | $O(n^2) \cdot (\# \text{ of iterations})$<br><b>Fast</b>                                   | $\approx O(n^2)$<br>+ accuracy,<br>matrix params<br>etc.         |
| Accuracy: | Exact<br><u><math>\log \log(1/\epsilon)</math></u>  | $\log(1/\epsilon)$<br><u><math>1/\epsilon</math></u> <u><math>1/\sqrt{\epsilon}</math></u> | <b>Faster (sometimes)</b><br>$\frac{1}{\epsilon^2}$ $1/\epsilon$ |

# SKETCHED REGRESSION

Randomized approximate regression using a Johnson-Lindenstrauss Matrix:



Input:  $A \in \mathbb{R}^{n \times d}$ ,  $b \in \mathbb{R}^n$ .

$$\tilde{x}^* = \arg \min_x \|\tilde{A}x - \tilde{b}\|_2^2$$

Goal: Want  $\min_x \|\tilde{A}x - \tilde{b}\|_2^2 \leq (1 + \epsilon) \min_x \|Ax - b\|_2^2$

to prove:  $\|A\tilde{x}^* - b\|_2^2 \leq (1 + \epsilon) \|A\tilde{x}^* - b\|_2^2$

Claim: Suffices to prove that for all  $x \in \mathbb{R}^d$ , where  $x^* = \arg \min_x \|Ax - b\|_2^2$

$$(6-6) \left( \|Ax - b\|_2^2 \leq \|\Pi Ax - \Pi b\|_2^2 \leq (1 + \epsilon) \|Ax - b\|_2^2 \right)$$

## Lemma (Distributional JL)

If  $\Pi$  is chosen to a properly scaled random Gaussian matrix, sign matrix, sparse random matrix, etc., with  $O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$  rows then for any fixed  $y$ ,

$$(1 - \epsilon) \|y\|_2^2 \leq \|\Pi y\|_2^2 \leq (1 + \epsilon) \|y\|_2^2$$

with probability  $(1 - \delta)$ .

$$y = Ax - b$$

**Corollary:** For any fixed  $x$ , with probability  $(1 - \delta)$ ,

$$(1 - \epsilon) \|Ax - b\|_2^2 \leq \|\Pi Ax - \Pi b\|_2^2 \leq (1 + \epsilon) \|Ax - b\|_2^2.$$

## FOR ANY TO FOR ALL

How do we go from "for any fixed  $\mathbf{x}$ " to "for all  $\mathbf{x} \in \mathbb{R}^d$ ".

This statement requires establishing a Johnson-Lindenstrauss type bound for an infinity of possible vectors  $(\mathbf{Ax} - \mathbf{b})$ , which obviously can't be tackled with a union bound argument.

$x_1, x_2, \dots, x_{100}, \dots$

$$O\left(\frac{\log 1/\delta}{\epsilon^2}\right)$$

Then for each  $x_i$

$$\|\mathbf{Ax}_i - \mathbf{b}\|_2^2 = (1 \pm \epsilon) \|\mathbf{A}\mathbf{x}_i - \mathbf{b}\|_2^2$$

with prob.  $1 - \delta$ ,

so with prob  $1 - 100\delta$  this holds  
for  $x_1, \dots, x_{100}$

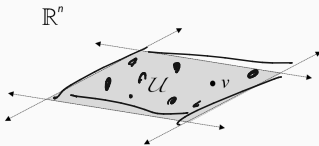
## SUBSPACE EMBEDDINGS

### Theorem (Subspace Embedding from JL)

Let  $\mathcal{U} \subset \mathbb{R}^n$  be a  $d$ -dimensional linear subspace in  $\mathbb{R}^n$ . If  $\Pi \in \mathbb{R}^{m \times d}$  is chosen from any distribution  $\mathcal{D}$  satisfying the Dist. JL Lemma, then with probability  $1 - \delta$ ,

$$(1 - \epsilon) \|\mathbf{v}\|_2 \leq \|\Pi \mathbf{v}\|_2 \leq (1 + \epsilon) \|\mathbf{v}\|_2$$

for all  $\mathbf{v} \in \mathcal{U}$ , as long as  $m = O\left(\frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon^2}\right)^1$ .



<sup>1</sup>It's possible to obtain a slightly tighter bound of  $O\left(\frac{d + \log(1/\delta)}{\epsilon^2}\right)$ . It's a nice challenge to try proving this.

## SUBSPACE EMBEDDING TO APPROXIMATE REGRESSION

**Corollary:** If we choose  $\Pi$  and properly scale then, with  $O(d/\epsilon^2)$  rows, then with high probability,

$$(1 - \epsilon) \|Ax - b\|_2^2 \leq \|\Pi Ax - \Pi b\|_2^2 \leq (1 + \epsilon) \|Ax - b\|_2^2$$

for all  $x$  and thus

for any  $x$  we can be written as  $Ax - b$ , want  $(1 - \epsilon) \|y\|_2^2 \leq \|\Pi y\|_2^2 \leq (1 + \epsilon) \|y\|_2^2$

$$\min_x \|\Pi Ax - \Pi b\|_2^2 \leq (1 + \epsilon) \min_x \|Ax - b\|_2^2.$$

i.e. we can solve linear regression approximately using the  $O(d/\epsilon^2) \times d$  matrix  $\Pi A$  in place of  $A$ .

**Proof:** Apply Subspace Embedding Thm. to the  $(d + 1)$  dimensional subspace spanned by  $A$ 's  $d$  columns and  $b$ . Every vector  $Ax - b$  lies in this subspace.

## SUBSPACE EMBEDDINGS

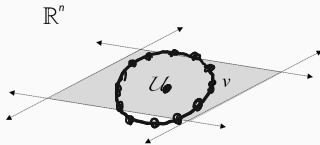
### Theorem (Subspace Embedding from JL)

Let  $\mathcal{U} \subset \mathbb{R}^n$  be a  $d$ -dimensional linear subspace in  $\mathbb{R}^n$ . If  $\Pi \in \mathbb{R}^{m \times d}$  is chosen from any distribution  $\mathcal{D}$  satisfying the Dist. JL Lemma, then with probability  $1 - \delta$ ,

$$(1 - \epsilon)\|\mathbf{v}\|_2 \leq \|\Pi\mathbf{v}\|_2 \leq (1 + \epsilon)\|\mathbf{v}\|_2$$

(1)

for all  $\mathbf{v} \in \mathcal{U}$ , as long as  $m = O\left(\frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon^2}\right)^2$ .



<sup>2</sup>It's possible to obtain a slightly tighter bound of  $O\left(\frac{d + \log(1/\delta)}{\epsilon^2}\right)$ . It's a nice challenge to try proving this.

## SUBSPACE EMBEDDING PROOF

**Observation:** The theorem holds as long as (1) holds for all  $\mathbf{w}$  on the unit sphere in  $\mathcal{U}$ . Denote the sphere  $S_{\mathcal{U}}$ :

$$S_{\mathcal{U}} = \{\mathbf{w} \mid \underline{\mathbf{w} \in \mathcal{U}} \text{ and } \underline{\|\mathbf{w}\|_2 = 1}\}.$$

**Follows from linearity:** Any point  $\mathbf{v} \in \mathcal{U}$  can be written as  $c\mathbf{w}$  for some scalar  $c$  and some point  $\mathbf{w} \in S_{\mathcal{U}}$ .

- If  $\underline{(1 - \epsilon)\|\mathbf{w}\|_2 \leq \|\Pi\mathbf{w}\|_2 \leq (1 + \epsilon)\|\mathbf{w}\|_2}$ .
- then  $c(1 - \epsilon)\|\mathbf{w}\|_2 \leq c\|\Pi\mathbf{w}\|_2 \leq c(1 + \epsilon)\|\mathbf{w}\|_2$ ,
- and thus  $(1 - \epsilon)\|c\mathbf{w}\|_2 \leq \|\Pi c\mathbf{w}\|_2 \leq (1 + \epsilon)\|c\mathbf{w}\|_2$ .

$\downarrow$   
 $\mathbf{v}$

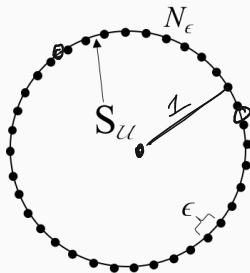
$\downarrow$   
 $\mathbf{v}$

$\downarrow$   
 $\mathbf{v}$

$c = \frac{1}{\|\mathbf{w}\|_2}$   
 $\mathbf{v} = \|\mathbf{v}\|_2 \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|_2}$   
 $\mathbf{w} \in S_{\mathcal{U}}$

## SUBSPACE EMBEDDING PROOF

**Intuition:** There are not too many “different” points on a  $d$ -dimensional sphere:



$$\underline{O\left(\frac{1}{\epsilon}\right)}$$

$$O\left(\frac{1}{\epsilon^2}\right)$$

$$O\left(\frac{1}{\epsilon^d}\right)$$

$N_\epsilon$  is called an “ $\epsilon$ ”-net.

If we can prove

$$\|w\| (1 - \epsilon) \leq \|\Pi w\|_2 \leq (1 + \epsilon) \|w\|_r$$

for all points  $w \in \underline{N_\epsilon}$ , we can hopefully extend to all of  $S_U$ .

## Lemma ( $\epsilon$ -net for the sphere)

For any  $\epsilon \leq 1$ , there exists a set  $N_\epsilon \subset S_{\mathcal{U}}$  with  $|N_\epsilon| = \left(\frac{4}{\epsilon}\right)^d$  such that  $\forall \mathbf{v} \in S_{\mathcal{U}}$ ,

$$\min_{\mathbf{w} \in N_\epsilon} \|\mathbf{v} - \mathbf{w}\| \leq \epsilon.$$

Want (1)  $\rightarrow (1/4) \|\mathbf{w}\|_2^2 \leq \|\Pi \mathbf{w}\|_2^2 \leq \|\mathbf{w}\|_2^2 \cdot (1+\epsilon)$   
to hold for all  $\mathbf{w} \in N_\epsilon$ .

$$\text{Set } \delta = \frac{1}{|N_\epsilon|} \quad \log(1/\delta) \quad \log(|N_\epsilon|)$$

$$= \log\left(\frac{4}{\epsilon}\right)^d$$

$$= d \log(4/\epsilon)$$

1. Set  $\delta = \left(\frac{\epsilon}{8}\right)^d$ . By a union bound, with high probability, for all  $\mathbf{w} \in N_\epsilon$ ,

$$(1 - \epsilon) \leq \|\Pi \mathbf{w}\|_2 \leq (1 + \epsilon).$$

as long as  $\Pi$  has  $O\left(\frac{\log(1/\delta)}{\epsilon^2}\right) = O\left(\frac{d \log(1/\epsilon)}{\epsilon^2}\right)$  rows.

2. Consider any  $\mathbf{v} \in S_{\mathcal{U}}$ . You can check that, for some  $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2 \dots \in N_{\epsilon}$ ,  $\mathbf{v}$  can be written:

$$\mathbf{v} = \mathbf{w}_0 + c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + \dots$$

for constants  $c_1, c_2, \dots$  where  $|c_i| \leq \epsilon^i$ .

3. Applying triangle inequality, we have

$$\begin{aligned} \|\Pi \mathbf{v}\|_2 &= \|\Pi \mathbf{w}_0 + c_1 \Pi \mathbf{w}_1 + c_2 \Pi \mathbf{w}_2 + \dots\| \\ &\leq \|\Pi \mathbf{w}_0\| + \epsilon \|\Pi \mathbf{w}_1\| + \epsilon^2 \|\Pi \mathbf{w}_2\| + \dots \\ &\leq (1 + \epsilon) + \epsilon(1 + \epsilon) + \epsilon^2(1 + \epsilon) + \dots \\ &\leq 1 + O(\epsilon). \end{aligned}$$

4. Similarly,

$$\begin{aligned}
 \|\Pi v\|_2 &= \|\Pi w_0 + c_1 \Pi w_1 + c_2 \Pi w_2 + \dots\| \\
 &\geq \|\Pi w_0\| - \epsilon \|\Pi w_1\| - \epsilon^2 \|\Pi w_2\| - \dots \\
 &\geq (1 - \epsilon) - \epsilon(1 + \epsilon) - \epsilon^2(1 + \epsilon) - \dots \\
 &\geq 1 - O(\epsilon).
 \end{aligned}$$

So we have proven

$$1 - O(\epsilon) \leq \|\Pi v\|_2 \leq 1 + O(\epsilon)$$

for all  $v$  in  $S_{\mathcal{U}}$ .

Adjusting  $\epsilon$  proves the Subspace Embedding theorem.

[t]

**Theorem (Randomized Linear Regression)**

Let  $\Pi$  be a properly scaled JL matrix (random Gaussian, sign, sparse random, etc.) with  $O\left(\frac{d \log(1/\epsilon)}{\epsilon^2}\right)$  rows. Then with high probability, for any  $\mathbf{A} \in \mathbb{R}^{n \times d}$  and  $\mathbf{b} \in \mathbb{R}^n$ ,

$$\min_{\mathbf{x}} \|\Pi \mathbf{A} \mathbf{x} - \Pi \mathbf{b}\|_2^2 \leq (1 + \epsilon) \min_{\mathbf{x}} \|\mathbf{A} \mathbf{x} - \mathbf{b}\|_2^2$$

